

Ting-Chih Chen

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Education

Virginia Tech

M.Sc in Computer Science and Applications

Blacksburg, VA

Jan. 2022 – Dec. 2023

National Changhua University of Education

B.Sc in Computer Science and Information Engineering

Changhua, TW

Sep. 2015 – June 2019

Experience

Contract ML Engineer

Ret[AI]ling Data | 🔗

Mar. 2025 – Present

New Taipei City, TW

- Developed a recommendation system using CLIP and Ollama for the retail industry, improving product relevance and enhancing customer shopping experiences in malls. 🔗
- Implemented a CDP using RAG, RESTful API, FastAPI, and MongoDB, enabling real-time tracking of user activity and generating insights into customer behavior.
- Documented API design guidelines, enabling company to efficiently replicate and scale the solution for new use cases.
- Utilized Selenium and BeautifulSoup to crawl dynamic, large-scale online retail data, extracting real-time product and market insights for analysis and strategic planning.

AI/ML Research Assistant

Virginia Tech

Mar. 2022 – Feb. 2025

Blacksburg, VA

- Developed a multimodal news claim generator using AMRBART. 🔗
- Developed an evidence summarization model with LLaMA2 and DeBERTaV3 using RLHF and PPO, achieving 48.6% accuracy and surpassing the SOTA model by 4.6%. 🔗
- Engineered a video dataset for inductive and abductive reasoning, leading data collection, cleaning, and ETL pipelines development.

Research Engineer Intern

National Center for High-Performance Computing

May 2019 – Aug. 2019

Taichung, TW

- Contributed in designing the highly secured dataset for the object detection task.
- Collaborated on ETL pipelines, data warehousing, and architecture, reducing processing time by 20%.
- Implemented YOLOv3 for the object detection task, achieving 82% mAP and surpassing the previous method by 4%.

Projects

When Trust is a Factor | 🔗

Mar. 2022 – Feb. 2025

- Engineered the multimodal data into a knowledge graph format using NetworkX, AMR and penman.
- Developed a multimodal summarization model using RLHF and PPO for fact-checking task.
- Developed a fine-grained inconsistency detector for multimodal multi-document information.

Publications

- Tang, C.-W., Chen, T.-C., Nguyen, K. A., Mehrab, K. S., Ishmam, A. M., & Thomas, C. M3D: MultiModal MultiDocument Fine-Grained Inconsistency Detection. In Proceedings of the **2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)**, November 12-16, 2024, Miami, USA (pp. 22270–22293). 📄
- Chen, T.-C., Tang, C.-W., & Thomas, C. MetaSumPerceiver: Multimodal Multi-Document Evidence Summarization for Fact-Checking. In Proceedings of the **62nd Annual Meeting of the Association for Computational Linguistics (ACL)**, August 11–16, 2024, Bangkok, Thailand (pp. 8742–8757). 📄
- Chen, T.-C.. Multimodal Multi-Document Evidence Summarization For Fact-Checking. **Master's thesis**, December 1, 2023, Virginia Tech. 📄

- Chen, T.-C., & Lee, C.-I. Application of LSTM Neural Network in Stock Price Movement Forecasting with Technical Analysis Index. In Proceedings of the 17th **International Conference on Innovation and Management (IAM)**, February 4-7, 2020, Sapporo, Japan (pp. 85-92). 

Awards

National Center for High-Performance Computing

Research Intern Award

Taichung, TW

Aug. 2019

National Changhua University of Education

Senior Project Award

Changhua, TW

Jun. 2019

Academic Service

Reviewer

- ACL Student Research Workshop (SRW) 2025.
- ACM Transactions on Intelligent Systems and Technology (TIST) 2024.

Teaching

CS5024 Ethics and Professionalism in CS

Teaching Assistant

Blacksburg, VA

Jan. 2023 - May 2019

CS3114 Data Structures and Algorithms

Teaching Assistant

Blacksburg, VA

Aug. 2022 - Dec. 2022

Skills

- **Programming:** Python, Java, C, C++, Javascript, HTML, CSS, php, SQL, R
- **Technologies:** PyTorch, Scikit-learn, Tensorflow, Hugging Face, OpenCV, ROS
- **Database:** MongoDB, Redis, MySQL
- **Platforms:** AWS, GCP

References

Chris Thomas

CS Assistant Professor @ Virginia Tech

Blacksburg, VA

- Email: chris@cs.vt.edu
- Website: <https://people.cs.vt.edu/chris/>
- Relationship: Master's advisor

Ismiini Lourentzou

iSchool Assistant Professor @ UIUC

Champaign, IL

- Email: lourent2@illinois.edu
- Website: <https://ischool.illinois.edu/people/ismini-lourentzou>
- Relationship: Member of Master's thesis committee